

## Project Title: Shopee App Review Analysis

**Disclaimer:** This is a personal project created solely for educational and portfolio purposes. The data used is a public dataset obtained from [Kaggle](#) and reflects only a specific time period. This dashboard is not affiliated with, sponsored by, or representative of the internal data or official views of Shopee or any related companies.

## Background

The e-commerce industry is fundamentally driven by continuous digital evolution, requiring businesses to consistently evaluate and enhance their platforms to remain competitive. For major e-commerce platforms like Shopee, delivering a seamless digital user experience is a critical determinant of long-term success and market dominance.

A significant factor influencing customer trust and user acquisition is the public perception shaped by app reviews on platforms such as the Google Play Store. Evaluating these reviews is essential, as an application's public rating directly impacts the brand's credibility and its ability to attract new customers. Without a structured evaluation of this feedback, companies may miss crucial opportunities to address service shortcomings and capitalize on their strengths.

App review analytics plays a vital role in transforming massive volumes of unstructured user feedback into actionable insights. By deeply analyzing this qualitative data, businesses can provide stakeholders with the necessary perspectives to make informed, data-driven decisions, prioritize service improvements, and elevate the overall customer experience.

Therefore, this project aims to analyze Shopee's Google Play Store app reviews to extract valuable business insights and identify specific areas for service enhancement. The analysis will classify user feedback into eight distinct categories: Shipping & Logistics, App Performance, Promotions & Coupons, Payments, Customer Service / Seller Support, General Praise, General Complaints, and Others. Furthermore, this project will track historical complaint trends on a monthly and yearly basis, alongside evaluating the percentage distribution of 1 to 5-star ratings. Ultimately, the results of this analysis are intended to guide strategic improvements, boost the application's overall rating, and foster greater customer trust to drive new user acquisition.

## Project Selection Criteria

Before starting the analysis process, several criteria were established to ensure that the project would provide meaningful insights, maintain data credibility, and simulate real-world data analytics practices. The project selection was based on the following considerations:

1. The dataset must be publicly accessible and not contain private or confidential information.
2. The analysis object should represent an industry or topic that is widely recognized and easy to understand by general audiences.
3. The data source must come from trusted and credible platforms to ensure that the validity and reliability of the secondary data can be justified.
4. The project documentation and workflow will follow the Google Data Analytics framework, consisting of Ask, Prepare, Process, Analyze, Share, and Act.

## Business Problem

Shopee receives thousands of app reviews on the Google Play Store, making manual analysis highly inefficient and impractical. The complexity increases significantly when a single user review addresses multiple areas for improvement across different service categories. Due to this overwhelming volume and complexity of unstructured data, there is a critical need to develop an intuitive, easy-to-understand dashboard or report. This solution will enable stakeholders, including both management and IT teams, to quickly grasp key insights and make data-driven decisions to resolve user issues.

## Objective

The primary goals of this analysis are to extract actionable insights from customer feedback to guide service improvements. Specifically, the objectives are:

1. **Identify Priority Areas for Improvement:** To determine which specific service categories (e.g., Shipping, App Performance, Customer Service) receive the most negative feedback and require immediate enhancement.
2. **Analyze Complaint Trends Over Time:** To track and quantify the volume of customer complaints across different months and years, revealing seasonal patterns or the impact of recent app updates.
3. **Determine Rating Distribution:** To calculate the percentage breakdown of customer ratings (from 1-star to 5-star) to understand overall user satisfaction levels and gauge general sentiment.

## Dataset

The data utilized for this project was sourced from Kaggle and originates from user reviews of the Shopee application on the Indonesian Google Play Store. You can access the original dataset here: [Shopee App Reviews from Google Play Store](#).

The raw dataset comprises 85,500 records containing user reviews written in Indonesian. The dataset structure includes the following columns:

1. **reviewId:** A unique identifier assigned to each user review.
2. **userName:** The display name of the user who submitted the review.
3. **userImage:** The URL linking to the profile picture of the reviewer.
4. **content:** The main text body containing the user's actual review and feedback.
5. **score:** The quantitative rating provided by the user, measured on a standard scale of 1 to 5 stars.
6. **thumbsUpCount:** The total number of "helpful" or "thumbs up" votes the review received from other users.
7. **reviewCreatedVersion:** The specific version of the application installed when the user wrote the review (Note: contains missing values).
8. **at:** The exact date and timestamp indicating when the review was published.
9. **replyContent:** The developer's response to the user's review (Note: This column is entirely empty in the current dataset).
10. **repliedAt:** The date and timestamp of the developer's response (Note: This column is also empty).

11. **appVersion**: The version of the application currently associated with the user's review (Note: contains missing values).

## Tools Used

To effectively process the data and build the final interactive dashboard, the following tools and technologies were utilized:

- **Python & Pandas**: Used for programmatic data manipulation, text processing, and data cleaning.
- **Spreadsheet (Google Sheets/Excel)**: Served as the primary data storage medium and integration layer to bridge the processed data with the visualization tool.
- **Google Looker Studio**: Employed to develop the interactive dashboard, creating accessible and comprehensive data visualizations for stakeholders.

## Methodology

This project follows the Google Data Analytics framework. Each stage was documented to simulate a real-world data analytics workflow, ensuring a structured and comprehensive approach to solving the business problem. The methodology consists of six key phases:

1. **Ask**: Defining the primary business problem, understanding stakeholder objectives, and formulating the key questions regarding app performance and user satisfaction.
2. **Prepare**: Gathering the dataset from Kaggle, understanding its structure, and assessing data integrity to ensure it is relevant and sufficient for the analysis.
3. **Process**: Cleaning and transforming the raw data using Python and Pandas. This includes handling missing values, standardizing text, and preparing the review content for categorization.
4. **Analyze**: Examining the processed data to identify patterns and trends. In this phase, user reviews are categorized into the seven predefined business areas, and rating distributions and temporal trends are calculated.
5. **Share**: Creating compelling data visualizations and building an interactive dashboard in Google Looker Studio to effectively communicate the insights to stakeholders.
6. **Act**: Formulating actionable, data-driven recommendations based on the findings to help Shopee management prioritize service improvements and enhance user experience.

## Process

The execution phase was divided into three primary stages: Data Cleaning, Data Transformation, and Exploratory Data Analysis (EDA).

### A. Data Cleaning

To ensure the integrity and accuracy of the analysis, the raw dataset underwent rigorous cleaning:

1. **Handling Inconsistencies & Nulls**: Removed entries that did not align with the established schema, as well as rows containing entirely empty review data.
2. **Deduplication**: Identified and removed duplicate records to prevent skewed metrics.
3. **Date Formatting**: Extracted and reformatted the `at` column into standard Month and Year formats to facilitate accurate time-series analysis.

## B. Data Transformation

To extract meaningful insights, the data was enriched and transformed through the following steps:

1. **Sentiment Classification:** Created a new "Sentiment" column to classify user feedback based on their given score. Reviews with **1 to 3 stars** were labeled as *Negative Sentiment*, while **4 to 5 stars** were categorized as *Positive Sentiment*.
2. **Keyword-Based Text Categorization:** Implemented a text-parsing logic to classify each review into one of **8 specific categories** based on predefined Indonesian keywords:
  - **Shipping & Logistics:** Reviews focusing on shipping services, couriers, or delivery times. (Keywords: *kurir, paket, lama, ekspedisi, pengiriman, resi, lacak, nyasar, belum sampai, hilang, j&t, spx, shopee express, jne, ongkir, lelet*)
  - **App Performance:** Complaints regarding bugs, errors, or technical system issues. (Keywords: *loading, lemot, ngelag, lag, error, bug, upgrade, update, force close, berat, jaringan, sinyal, keluar sendiri, macet, apk, aplikasi*)
  - **Promotions & Coupons:** Comments regarding events (e.g., 12.12), voucher availability, free shipping, and product pricing. (Keywords: *voucher, voucher, gratis ongkir, promo, diskon, event, flash sale, mahal, koin, minimal belanja, min belanja, 12.12, ongkos kirim, klaim*)
  - **Payments:** Feedback on Shopee's financial features such as ShopeePay, SPayLater, transfers, and refunds. (Keywords: *spaylater, paylater, shopeepay, kredit, transfer, top up, saldo, refund, pengembalian dana, gagal bayar, tagihan, bunga, denda, rekening*)
  - **Customer Service / Seller Support:** Interactions with Shopee Customer Service (CS) or individual sellers within the app. (Keywords: *cs, customer service, admin, bot, penjual, seller, toko, respon, penipuan, komplain, retur, kembalikan barang, kecewa pelayanan*)
  - **General Praise:** Positive reviews offering praise without citing specific features. (Keywords: *bagus, mantap, terbaik, good, membantu, suka, memuaskan, keren, oke, lanjutkan, bermanfaat*)
  - **General Complaints:** Reviews from frustrated users expressing general dissatisfaction or threatening to delete the app without specifying a distinct issue. (Keywords: *kecewa, hapus, uninstall, uninstal, buruk, jelek, nyesel, pindah, aplikasi sebelah, sampah*)
  - **Others:** A catch-all category for reviews that did not match any of the keywords from the seven primary categories above.
3. **Metric Aggregation:**
  - Calculated the overall average rating score.
  - Aggregated the total volume of complaints on an annual basis.
  - Computed the overall percentage of positive sentiment across the dataset.

## C. Exploratory Data Analysis (EDA)

With the data cleaned and categorized, analytical exploration was conducted to uncover key business insights:

1. **Trend Analysis:** Analyzed the historical year-over-year volume of complaints to identify spikes and shifts in user dissatisfaction.
2. **Rating Distribution:** Evaluated the percentage breakdown of each score (1 to 5 stars) to gauge overall application health.
3. **Category Prioritization:** Identified the specific service categories accumulating the highest volume of complaints, pinpointing the most urgent areas requiring business intervention.

## Key Findings

After analyzing the processed dataset, several significant patterns and insights emerged regarding user satisfaction and platform performance:

**1. Polarized Sentiment & Rating Distribution** The user sentiment is somewhat polarized. While positive sentiment holds a slight majority at **52.9%**, negative sentiment closely follows at **47.1%**.

- The breakdown of ratings reveals that **5-star reviews dominate at 43.9%**.
- However, **1-star reviews represent a critical 28.9%** of all feedback, indicating a large segment of highly dissatisfied users.
- The remaining ratings are distributed across 2-stars (8.8%), 3-stars (9.3%), and 4-stars (9.0%).

**2. Historical Complaint Volumes (2018 - 2024)** Out of 50,611 total recorded reviews spanning from 2018 to 2024, **23,838 reviews (47.1%)** were classified as complaints (negative sentiment). Historically, the top three categories driving user dissatisfaction are:

- **Shipping & Logistics:** The leading cause of complaints with **12,426** records.
- **App Performance:** Following closely with **11,350** records.
- **Customer Service / Seller Support:** The third most common issue with **5,686** records.

**3. Specific Trends in 2024** The year 2024 saw the highest volume of user feedback, recording 35,259 total reviews. Of these, **13,502** were complaints. When isolating the data to focus exclusively on 2024, a shift in user frustration points becomes apparent:

- **Shipping & Logistics** remained the primary pain point, accumulating **6,522** complaints.
- **App Performance** maintained its second-place position with **4,726** complaints.
- **General Complaints** surprisingly rose to become the third most frequent issue with **3,148** records, indicating a rise in general frustration without users citing specific technical or service issues.
- Additionally, temporal analysis revealed that the peak month for complaints during 2024 was **March**, which recorded **517** complaints.

## Recommendation

Based on the key findings, the following actionable recommendations are proposed for Shopee's management and technical teams:

1. **Optimize Shipping & Courier Operations:** Since "Shipping & Logistics" is consistently the primary driver of negative reviews, Shopee must heavily evaluate its logistics infrastructure (e.g., Shopee Express) and third-party partners. Implementing stricter Service Level Agreement

(SLA) monitoring for couriers, improving real-time tracking transparency, and providing faster resolutions for lost or delayed packages are critical steps to reduce user frustration.

2. **Enhance App Stability & Performance:** With over 11,000 historical complaints regarding app performance, the engineering team needs to prioritize bug fixing, reducing app size (bloatware), and optimizing load times. Special focus should be placed on load testing servers prior to peak event months (like March) to prevent lagging and application crashes during major sales.
3. **Address the Surge in "General Complaints":** The sudden rise of generic frustration in 2024 indicates overall user fatigue or dissatisfaction with recent platform changes. Shopee should conduct deeper User Experience (UX) research to identify hidden friction points and launch re-engagement or loyalty campaigns to prevent user churn to competitors.

## Conclusion

This project successfully demonstrates the value of data analytics in transforming unstructured consumer feedback into strategic business insights. By analyzing over 50,000 processed reviews, we identified that while Shopee maintains a strong base of highly satisfied users (43.9% giving 5 stars), critical operational bottlenecks, specifically in shipping logistics and app stability, are driving a significant volume of 1-star ratings (28.9%).

The developed Google Looker Studio dashboard now allows stakeholders to dynamically monitor customer sentiment and pinpoint exact areas of friction. By prioritizing the recommended improvements in delivery services, technical performance, and proactive customer engagement, Shopee can effectively reduce negative sentiment, elevate its overall app rating, and foster greater trust to drive sustainable user growth in the competitive e-commerce landscape.